



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

NAME	RIFAQ AJMAL
REG NO	23MDBC435
SECTION	A
DEPPT	COMPUTER SCIENCE
SUBJECT	DLD LAB
TEACHER	SHEHZAD SIR

LAB TASK : 2

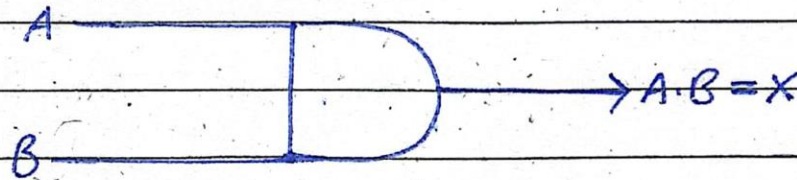
And Gate

It is logical gate
their execute the logical multi-
plication of binary inputs.
→ It give output 1 when input
is 1, otherwise the rest will be
zero

Expression

$$A \cdot B = X$$

Diagram



Name

IC 7408

Truth Table

A	B	A · B
0	1	0
0	0	0
1	1	1
1	0	0

IC or connecting wires with the.

* first connect a wire with 1st and then pin of I.C.

* Second connect GND in n position.

* Take two wires and connect it with the (A1-AE) in the bread board pin (3) and (6)

Reaching

Pin A4	Pin A8	LEN7
0	0	0
1	0	0
0	1	0
1	1	1

Advantages

* it perform logical & conjunction operations. * It means all produced inputs will give output (1).

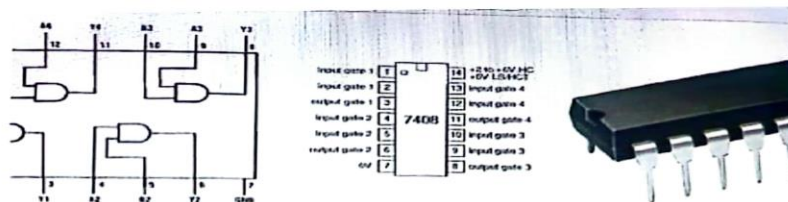
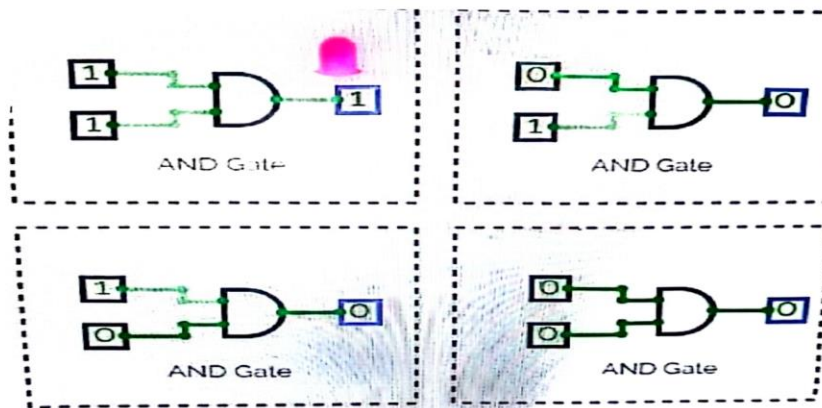
* It consumes low power as compare to other computer logic gates.

5) * It is very easy to improve use C.M.O.S and many other.

Disadvantage

* And gate has limitized range, this means it will give bioghly output when the input are height else when they give low output.

* It introduced the pre-position delay which hight lead to synthosisium.



Application of And Gate

- * It is used to design complex circuit such as adder and multiplier.
- * It is used to control logic highly system.
- * It can be used in security system and used for along and operating machinery.

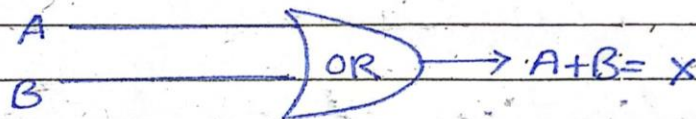
OR GATE

It is the common type of logical gates which provided its input high which output high otherwise low.

Expression

$$A + B = X$$

Diagram



Truth Table

A	B	A+B=X
0	0	0
0	1	1
1	0	1
1	1	1

5) Application of OR GATE

OR Gate is used in small doorbell which combining signal forms (doorbell) which combines signal.

Emergency lighting

OR Gate complex and combining some from multiple energy switches if any switch is high means turned on energy light.

Advantages

- * OR gate are said to be fundamental building blocks.
- * flexibility: OR Gate is used for its flexibility.
- * Simplicity: - OR gate is simple as compare to other gates.
- * Easy: - OR gate is simple and easy to make.

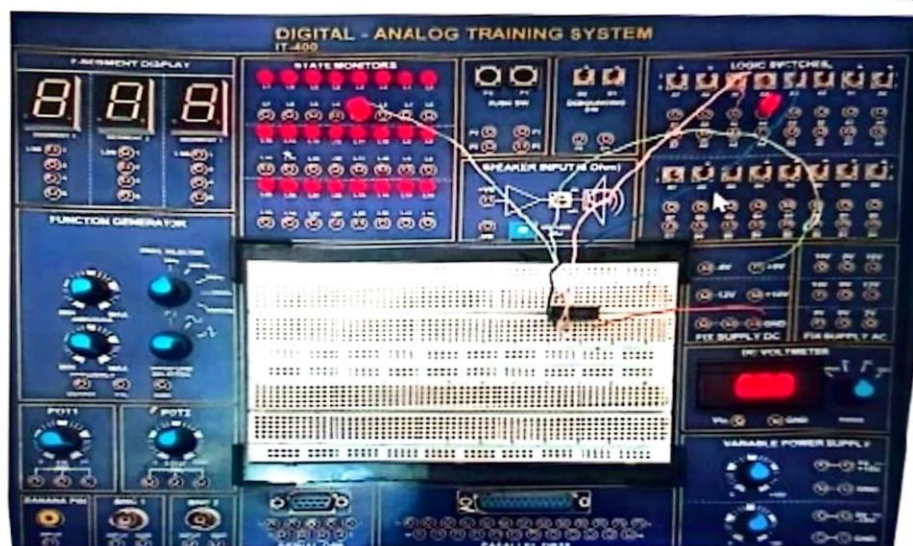
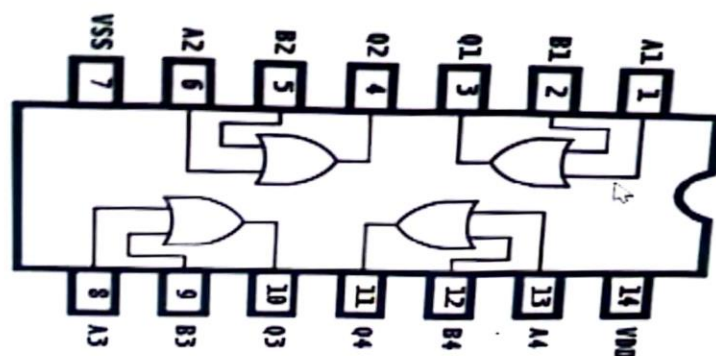
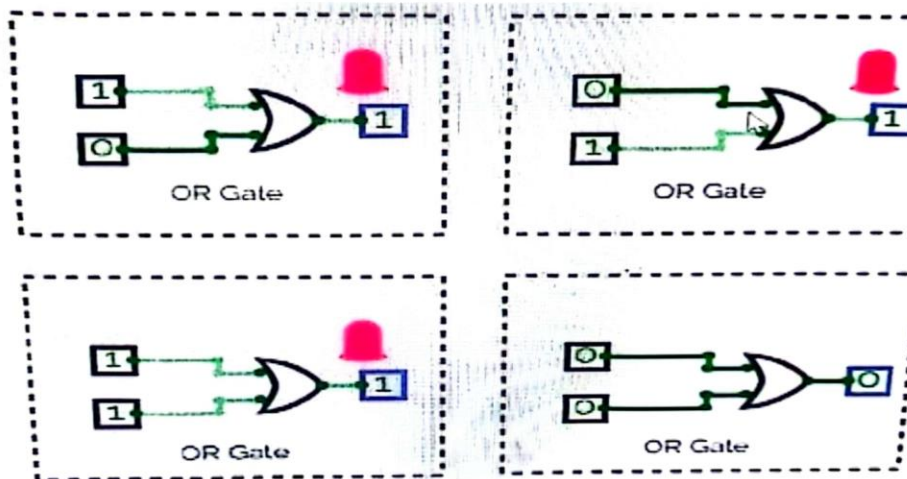
6) Disadvantage

* Lack of precision application that require precise revision moving based on individual input values:-

* Non invertable or gate is non-invertable its output is high at least input is highly otherwise low.

Reading Of OR Gate

Pin A1	Pin 4	LED 4
0	0	0
0	1	1
1	0	1
1	1	1



Reading Of OR Gate

Pin A1	Pin 4	LED 4
0	0	0
0	1	1
1	0	1
1	1	1

NAND GATE

It is a universal gate that can be used in boolean function without help of anything base gate.

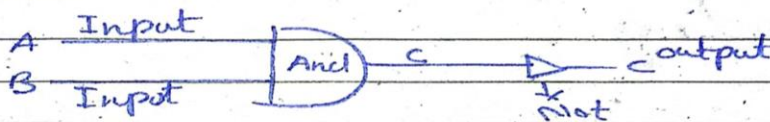
Composition

Nand gate is opposite to AND GATE. So when Not in Combines with And become Nand.

Name

7400 / 74000

Diagram



Truth Table

A	B	$A \cdot B$	$C = \overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Procedure

- First value two (AND) and not 0/4 moving it on bread board.

- * Taking wires and connecting it GND of both IC to the main trainer
- * Connect the both IC now taking another two wires and connect it into the two input.
- * Taking a small wire and connect to the output and its input to LED.

Reading

A _n	A _s	C _{ic}	Z
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

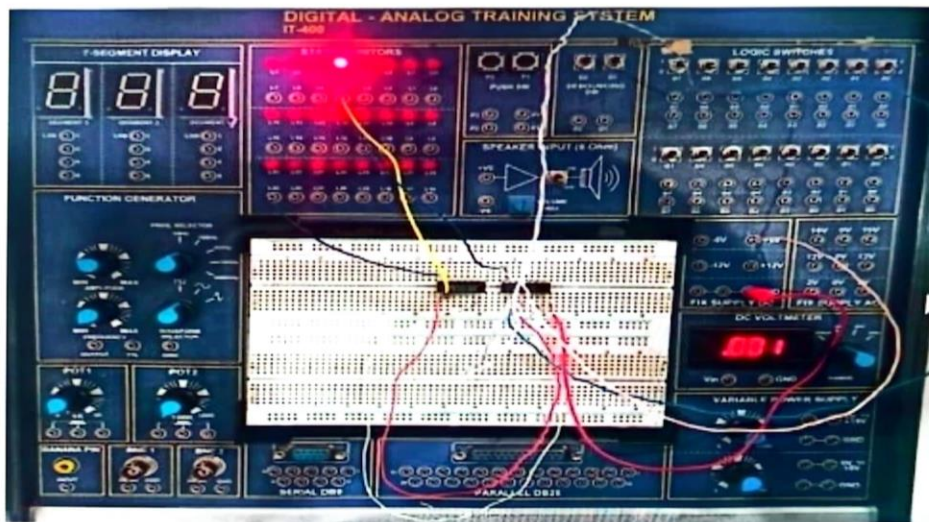
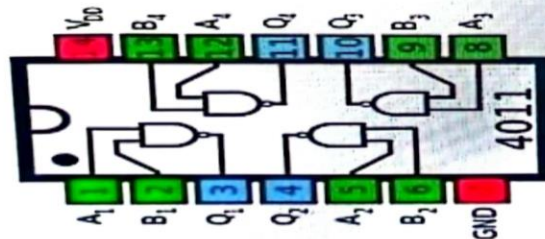
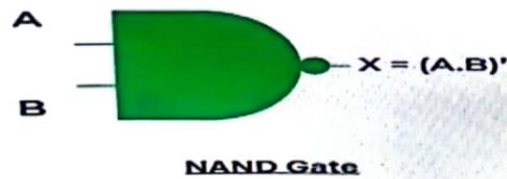
Application

- * Decodes And Encoder
- * Clock Generator
- * Many Users

9) Disadvantage

* It connect always the most optimized result by implementation (Nand Gate).

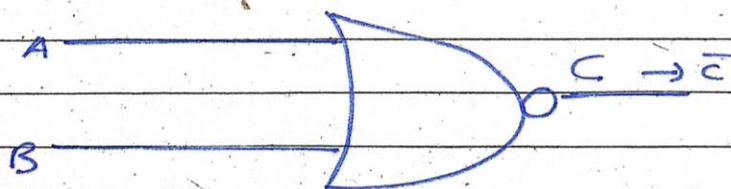
* Propagation delays some logical expression might needs some gates to resolve problem.



NOR Gate

Nor Gates are small digital logic gates that performs nor operator between two or more binary inputs and output binary signals.

Diagram



10) Truth Table

0	0	0	1
1	0	1	0
0	1	1	0
1	1	1	0

Composition of Nor Gate

- Trainer Board
- Bread Board
- Logical Switches
- LED
- Two input.
- One output

Reading

A	B	$C = A + B$	$C = \overline{A \cdot B}$
0	0	0	1
0	1	1	0
1	0	1	0

1) Steps

- Ist taking two IC's (00) ant (01) not and mainly if in the bread energy board.
- If the GND circuit the values date also proved IC of (00)(00)
- Now taking two wire and connect to the logical switches to the input of OR gate at pulse put (7) and output and circuit that input of not gate and Not Gate loops.

